higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY

(First Paper)

NQF LEVEL 2

(10401012)

21 February 2019 (X-Paper)
09:00–12:00

Calculators and drawing instruments may be used.

This question paper consists of 12 pages and 2 addenda.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly show ALL calculations, diagrams, graphs, and so forth that will be used in this paper.
5. Round off your answers to TWO decimals where necessary, unless otherwise stated.
6. Drawing instruments including rulers, pairs of compasses and protractors may be used.
7. Diagrams are not necessarily drawn to scale.
8. Answer QUESTIONS 4.2.5 on the attached ADDENDUM A and QUESTION 5.3 on the attached ADDENDUM B and hand these in with the ANSWER BOOK.
9. Write neatly and legibly.

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QUESTION 1

1.1 Calculate the following without using a calculator. Show ALL calculations.

1.1.1 $(6 + 20) - 8 \div 2$

1.1.2 $\frac{1}{2}$ of $200 - (-10)$

(2 × 3) (6)

1.2 Arrange the following numbers in descending order:

36; – 66; 0; 89 (2)

1.3 There are 186 students enrolled to write Mathematical Literacy Level 2. 102 of the students are females. What percentage of the students is female? Round the answer off to the nearest percentage. (3)

1.4 Convert 6 300 grams to kilograms. (2)

1.5 A laptop costs R10 200. How much would you pay for the laptop if you get a 15% discount? (3)

1.6 Divide R6 000 between Wandile and Gladys in the ratio 1 : 4. (4)

1.7 A runner trained for 3 days during the week. The following times were recorded during the week:



Monday: 1 hour 35 minutes

Wednesday: 1 hour 16 minutes

Friday: 2 hours 23 minutes

Add the times trained for the 3 days and calculate the total number of hours and minutes the runner trained for. (3)

1.8 Makro sells a case of 24 Coca-Cola cans for R162,95.

Pick and Pay sells a six-pack of Coca-Cola cans for R34,99.

Would it be cheaper to buy a case of 24 from Makro or the six pack from Pick and Pay?

Show ALL your calculations.





(4)

1.9 Write the digital time below in 12-hour notation (analogue time).



(1)

1.10 Write 0,32 as a percentage. (2)

[30]

QUESTION 2

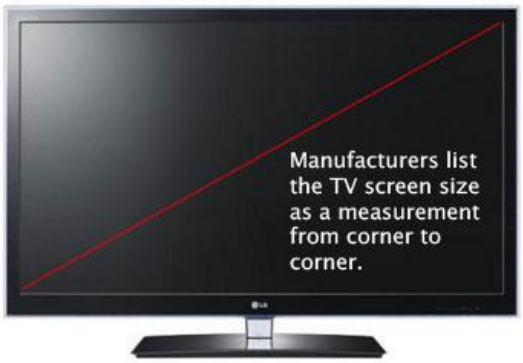
2.1 Choose a description from COLUMN B that matches a term/word in COLUMN A. Write only the letter (A–E) next to the question number (2.1.1–2.1.4) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1.1	Square	A	the distance around a circle
2.1.2	Hypotenuse	B	four equal sides and four right angles
2.1.3	Circumference	C	90-degree angle
2.1.4	Right angle	D	the longest side of a right-angled triangle, opposite the right angle
		E	the amount of space that a substance or object occupies

(4 × 2) (8)

2.2 The size of a television (TV) screen is determined by the diagonal measurement of the television screen as seen in the sketches below. Study the sketches and answer the questions that follow.





[*Source: http://www.tvguides.com*]

2.2.1 Paul bought a small television for his room. The viewable width of the TV is 28 inches and the viewable height of the TV is 15,7 inches.



Calculate the size of the television screen (in inches). Round your answer off to the nearest inch.

Formula: Size of screen = $\sqrt{\text{Viewable width}^2 + \text{Viewable height}^2}$ (5)

2.2.2 Use the dimensions in QUESTION 2.2.1 to calculate the area of the television screen.

Formula: Area = length × breadth (3)

2.3 Lufefe wants to buy a portable swimming pool. He found a picture of a portable swimming pool in a catalogue. The dimensions of the portable pool are:

Length = 412 cm
Breadth = 201 cm
Height = 122 cm



[Source: *http://www.gtmall.com.au*]

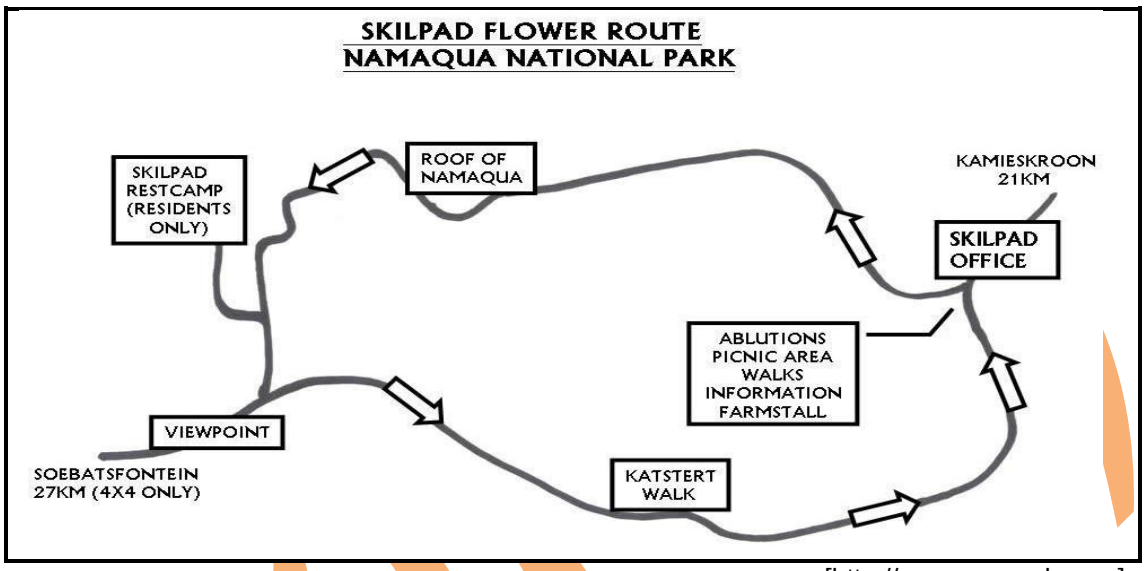
2.3.1 Calculate the volume of the portable swimming pool by using the measurements supplied in the picture above.

Formula: Volume = length × breadth × height (4)

2.3.2 Lufefe plans to put the portable swimming pool in the yard behind his house. The length of the yard is 4 m and the breadth is 3 m.

Will the portable swimming pool be able to fit into Lufefe's backyard? Justify your answer. (3)

2.4 Below is a simplified map of the Skilpad Flower Route in the Namaqua National Park. Study the map below and answer the questions that follow.



[http://www.sanparks.org]

2.4.1 The scale of the map is 1 : 50.

Calculate the actual distance of the Skilpad Flower Route if the distance on the map is 103 cm. Give your final answer in kilometres. (5)

2.4.2 Can any type of vehicle travel on the Soebatsfontein Road? Give a reason for your answer. (2)

[30]

QUESTION 3

3.1 Clark wants to upgrade his home theatre system. He received a quote from Maxisound Suppliers in Brakpan. Use the information to answer the questions.

MAXISOUND SUPPLIERS

64 Apple Road BRAKPAN 1541	☎ 011 915 0028 ✉ ben@maxisound.co.za

CUSTOMER	CLARK KENT		QUOTE NUMBER	Q12678
	21 ELM STREET		QUOTE DATE	01/11/2018
	BRAKPAN		SALES PERSON	BEN STOKES
	1541			

ALL QUOTES ARE VALID FOR 14 DAYS.

PRODUCT	ITEM CODE	QUANTITY	UNIT PRICE	TOTAL
JVC Bluetooth Speaker	JVC BS12	1	1 899,00	1899,00
JVC Surround Sound Speaker	JVC SSS45	6	115,00	690,00
Sony Stereo Amplifier	DH130	1	3299,00	3299,00
				5888,00

All prices are VAT inclusive (15%)

A discount of 5% is given on all cash payments.

3.1.1 Name the sales person who issued the quote? (1)

3.1.2 On which date was the quote issued? (1)

3.1.3 What is the item code for the Sony Stereo Amplifier? (1)

3.1.4 Show how the total of R690 was calculated for the JVC Surround Sound Speakers. (2)

3.1.5 Calculate the VAT exclusive amount of the JVC Bluetooth Speaker. (3)

3.1.6 Would the quote still be valid on 01/12/2018? Justify your answer. (2)

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3.2 Gatiep records his income and expenses every month. Below is the statement showing his income and expenditure from January to March 2018.

GATIEP'S INCOME AND EXPENSES FOR JANUARY TO MARCH 2018.			
INCOME	**JANUARY**	**FEBRUARY**	**MARCH**
Salary	5 000	5 000	5 000
Commission from sales	6 500	4 200	7 100
TOTAL INCOME	**11 500**	**9 200**	**A**
EXPENSES			
Rent	3 500	3 500	3 500
Car repayment	1 250	1 250	1 250
Insurance	700	700	700
Groceries	3 200	3 450	3 330
Cell phone	450	375	425
Entertainment	1 220	2 180	1 810
TOTAL EXPENSES	**10 320**	**11 455**	**11 015**
SURPLUS/DEFICIT	1 180	-2 255	**B**

Use the information to answer the questions.

3.2.1 Give ONE example of a variable income. (1)

3.2.2 Give ONE example of a fixed expense. (1)

3.2.3 Calculate the missing values for A and B on the income and expenditure statement. (2 × 2) (4)

3.2.4 Did Gatiep have a surplus or a deficit in January 2018? (1)

3.2.5 Gatiep had a deficit of R2 255 in February 2018. Give ONE example of an expense which caused this deficit in February. (1)

3.3 Regina hired a car for 2 days. She travelled a total distance of 545 kilometres. Use the information below to calculate how much it cost her to hire the car.

R465 per day
200 kms free per day thereafter
R3,85/km
Seats: 4
Air Con: Yes
Manual: Yes
Luggage: 2 big bags



(4)

3.4 The table below shows the adjusted bank fees of Standard bank from 2016 to 2017 for their Elite bank accountholders. Use the table to answer the questions.

Standard Bank

Standard Bank Elite Account	2016 Fees	2017 Fees	% Change from 2016 to 2017
Withdrawal (Native)	R4,00 + 1,20% of value	R1,80 per R100	-10,0%
Withdrawal (Other)	R6,70 + R4,00 + 1,20% of value	R6,70 + R1,80 per R100	-6,0%
Withdrawal (POS)	R5,00	R1,80	-64,0%
Deposit (ATM)	R4,00 + 1,30% of value	R1,0 per R100	-14,3%
Debit order (internal)	R4,20	R4,50	+7,1%
Debit order (external)	R13,10	R16,00	+22,1%
Account fee (PAYT)	R55,00	R55,00	–

[Source:https://businesstech.co.za]

3.4.1 What was the bank fee for a debit order (internal) in 2016? (1)

3.4.2 What was the percentage change for an ATM deposit from 2016 to 2017? (1)

3.4.3 What does the negative sign (–) mean in the % change from 2016 to 2017 column? (1)

3.4.4 How much did it cost to deposit R1 500 at an ATM in 2016? Show all calculations. (4)

3.4.5 In general, do you think that Standard Bank has increased or decreased their bank costs from 2016 to 2017? (1)

[30]

QUESTION 4

4.1 Study the following patterns and complete the missing numbers.

4.1.1 8 ; 16 ; 24 ; … ; … (2)

4.1.2 12 ; 24 ; 48 ; … ; … (2)

4.1.3 Is the pattern in QUESTION 4.1.1 an example of a constant difference or a constant ratio? (1)

4.2 It is estimated that it will take one cleaner an average of 50 hours to clean the classrooms at a college. The following table investigates the time it will take to clean the classrooms if there are more cleaners to help with the job.

Number of cleaners	1	2	5	**B**	25
Time taken in hours	50	25	**A**	5	2

Use the formula: Number of cleaners × Time = 50

4.2.1 Calculate the values for A and B in the table above. (2 × 2) (4)

4.2.2 Name the independent variable in the table above. (1)

4.2.3 Name the dependent variable in the table above. (1)

4.2.4 What is the relationship between the variables in the table above? Justify your answer. (3)

4.2.5 Use the table above to draw and label the graph on the grid found in ADDENDUM A (attached). Label the horizontal and vertical axes and provide a suitable heading for the graph. (8)

4.2.6 Use the graph in QUESTION 4.2.5 to approximate how long 15 cleaners will take to clean the classrooms? (1)

4.3 Andile sells handbags. The following table shows the income Andile will receive when she sells a certain number of handbags.



Number of handbags sold	0	1	5	**A**	15	20
Income in Rand	0	100	500	1000	**B**	2 000

Use the formula: Income in rand = Number of handbags sold × R100

4.3.1 What is the price of one handbag? (1)

4.3.2 How many handbags must Andile sell to have an income of R2 000? (1)

4.3.3 Calculate the values for A and B in the table above. (2 × 2) (4)

4.3.4 What is the relationship between the variables in the table above? (1)

[30]

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QUESTION 5

5.1 The Student Support Services (SSS) conducted a survey to determine the number of siblings (brothers and sisters) of each student at their college. The following table shows the results for the Tourism Level 2 students. Information was obtained from the registration forms that students completed at registration.

NUMBER OF SIBLINGS	TALLY	FREQUENCY
0	𝍸	5
1	𝍸 𝍸 ll	**A**
2	𝍸 𝍸 𝍸 𝍸 𝍸 ll	27
3	**C**	20
4	𝍸 lll	**B**
5	**D**	3
TOTAL		**E**

5.1.1 Did SSS make use of interviews, questionnaires or observations to get the above information? Justify your answer. (2)

5.1.2 Write down the missing frequency values for the letters A and B on the table. (2)

5.1.3 Write down the missing tally for the letters C and D on the table. (2)

5.1.4 Calculate E, the total number of Tourism Level 2 students at the college. (2)

5.1.5 What is the most frequent number of siblings? (1)

5.1.6 What percentage of the students has 5 siblings? (3)

5.2 Johannes Vetter was the winner of the Men's Javelin Throw at the recent IAAF World Championships that was held in London. Each competitor had 6 chances to throw the javelin. The following set of numbers represents the distance measured (in metres) for each of Johannes Vetter's throws:



88,26	91,20	87,52	86,38	85,44	90,02

5.2.1 Rearrange the set of distances in ascending order. (2)

5.2.2 A javelin measures 260 cm. Convert 260 cm to metres. (2)

5.2.3 Calculate the mean/average of the set of numbers. (3)

5.2.4 With which throw did he win the competition? (1)

5.2.5 The world record for javelin throwing is 104,8 metres.
How far off the record was Johannes's best throw? (2)

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5.3 The table below, shows the Mathematical Literacy L2 examination results of students at a campus:

Intervals	**Frequency**
0-19,9%	12
20-39,9%	20
40-59,9%	64
60-79,9%	21
80-100%	8

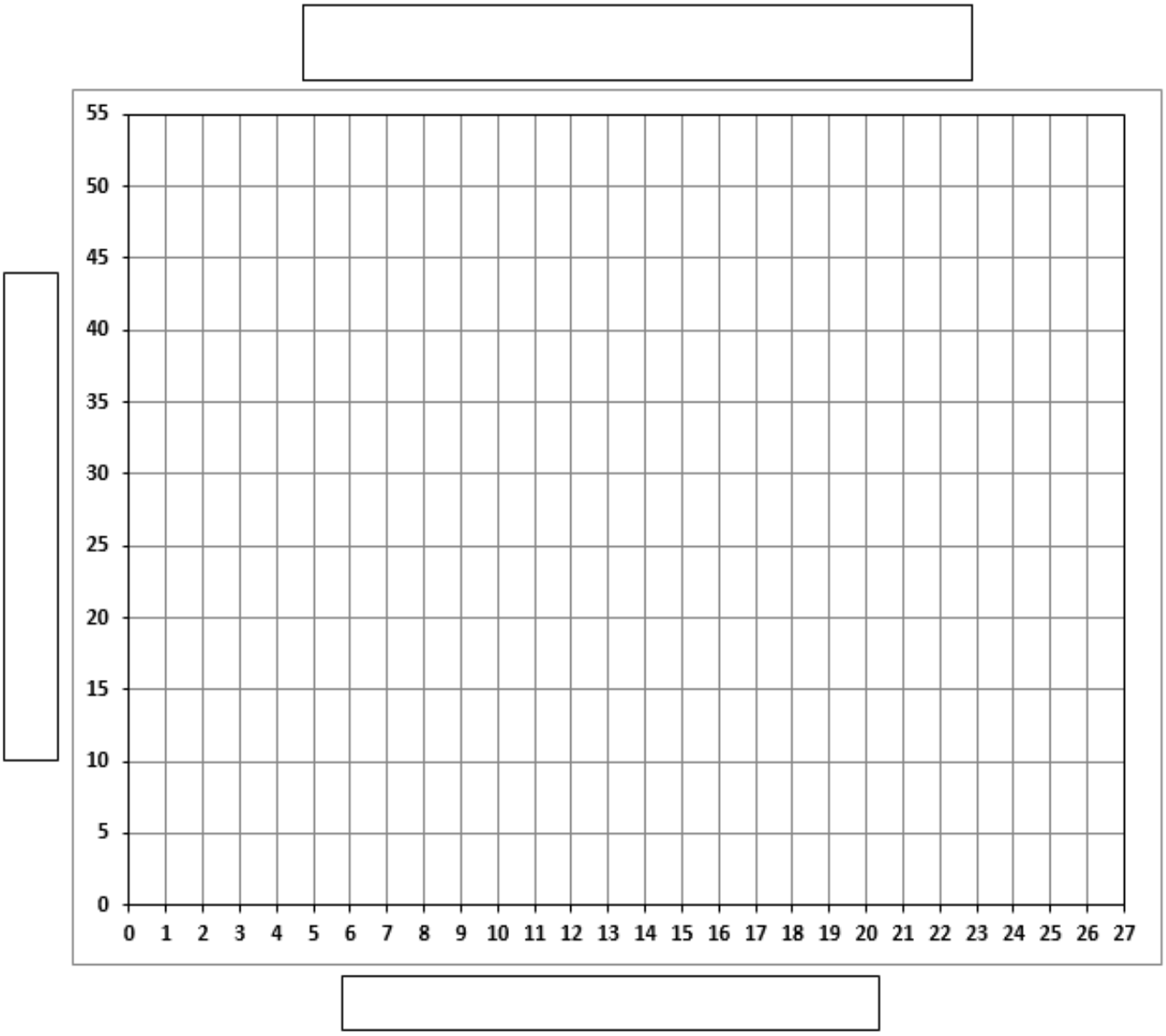
Draw and label a histogram of the examination results on the grid found on ADDENDUM B (attached). Provide a suitable heading for the graph and label the axes. (8)
[30]

TOTAL: 150

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ADDENDUM A **EXAMINATION NUMBER:**

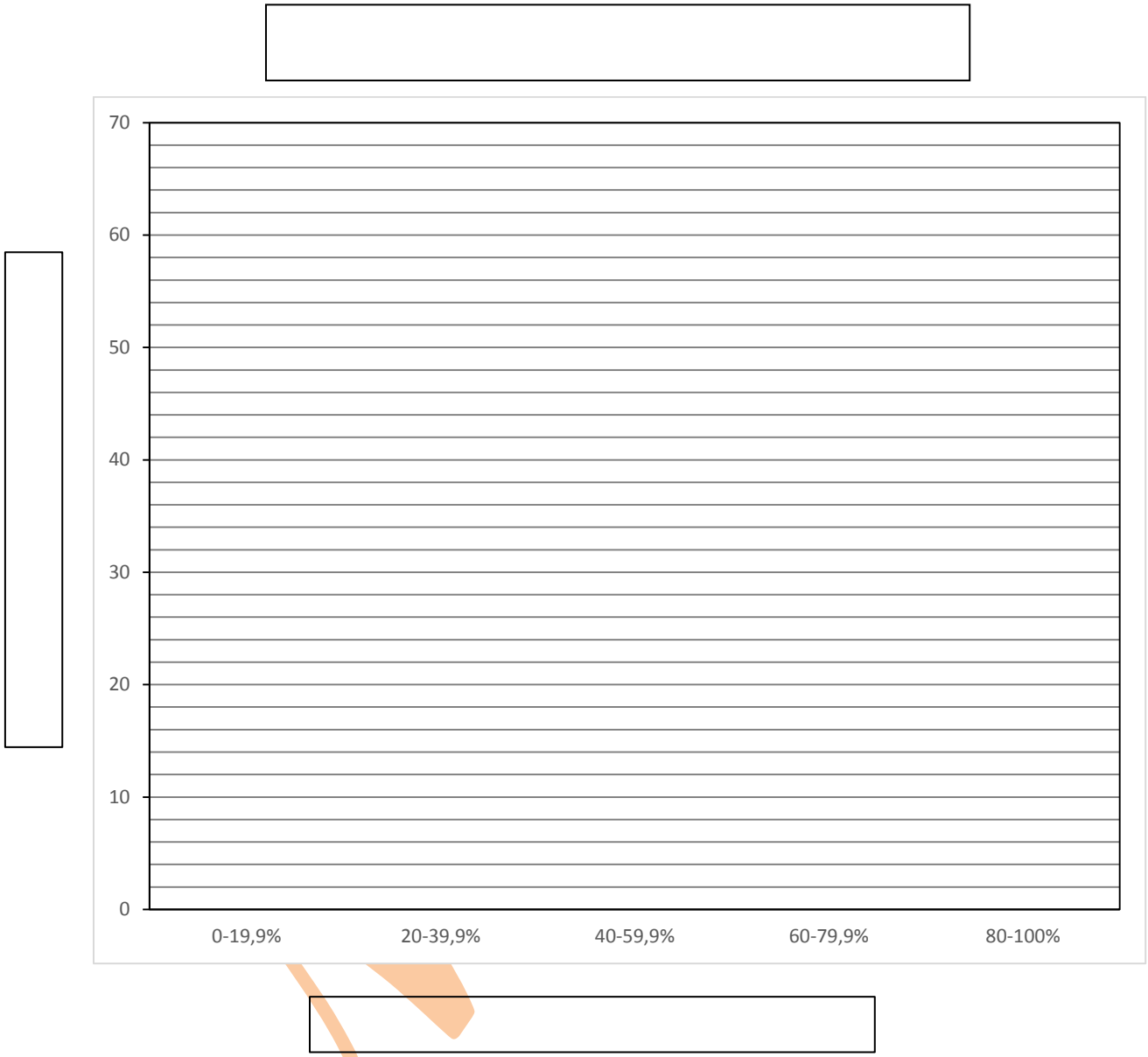
4.2.5



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ADDENDUM B **EXAMINATION NUMBER:**

5.3



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